NORTHERN ELECTRICAL TESTING COMPANY (NPCETC)

465 Nguyen Van Linh, Long Bien, Hanoi, Vietnam | Relay Testing Department

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Ha Noi, July 22, 2026

To: Ph.D. Selection Committee
Doctoral Program in Electrical, Electronics and Computer Engineering
College of Engineering and Computer Science (CECS)
University of Michigan–Dearborn, MI 48128, USA

Dear Admissions Committee,

We are writing to strongly recommend Xuan Thanh Trinh for admission to the Ph.D. program in Electrical, Electronics and Computer Engineering at the University of Michigan-Dearborn. We had the privilege of mentoring Thanh during his technical internship at the Relay Testing Department of the Northern Power Electric Testing Company (NPCETC) in the summer of 2025. While one of us ([Name of Head of Relay]) directly supervised his daily engineering analyses and hardware evaluations, the other ([Name of Vice Director]) oversaw his overarching technical deliverables and system reporting. Based on his exceptional analytical rigor and proven capacity to solve complex physical grid problems under strict industrial standards, we are confident that he is thoroughly prepared for the demands of doctoral research.

What distinguished Thanh from other interns was his remarkable ability to decode complex power grid systems purely through analytical observation, proving he possesses a highly robust technical foundation. First, since strict industrial safety protocols rightfully restricted him from directly operating high-voltage equipment, he focused his technical acumen on dissecting dense industrial schematics, specifically analyzing the intricate secondary circuits and protection protocols of the 110kV Tam Dao substation. Second, he demonstrated an extraordinary capacity to logically trace these electromechanical mechanisms in the field, accurately understanding how hardware interlock constraints directly dictate the configuration of numerical protective relays. This unique capability to mentally simulate and analyze real-world systems through blueprints and field observations proves he has the analytical foundation needed for rigorous, simulation-driven power system modeling.

Beyond his analytical capabilities, Thanh exhibited an innate drive for continuous learning and a high degree of professional maturity within our demanding corporate environment. First, he proactively shadowed our senior field engineers, displaying relentless intellectual curiosity by asking targeted questions that bridged his academic coursework with our practical, safety-critical field constraints. Second, he meticulously synthesized these field interactions and technical observations into a comprehensive internship report, demonstrating a level of technical communication and clarity rarely seen in undergraduate engineering students. His exceptional coachability, combined with his ability to absorb and articulate complex industrial concepts, guarantees that he will be an immediate, highly collaborative asset to your doctoral research community.

Ultimately, Thanh stands out as an exceptional engineer who seamlessly bridges robust analytical rigor with outstanding professional maturity and coachability. We have no doubt that his ability to navigate complex, real-world engineering constraints will make him an immediate and invaluable asset to your doctoral program. Therefore, we enthusiastically endorse his application for admission without hesitation. Should you require any further insights regarding his performance during his time with us, please do not hesitate to contact our office.

Sincerely,

[Name of Vice Director]
Vice Director
Northern Electrical Testing Company
Email: phogiamdoc@etc.com.vn

[Name of Head of Relay]
Head of Relay Testing Department
Northern Electrical Testing Company
Email: truongphong@npcetc.vn